

THIS IS A SOURCES SOUGHT NOTICE ONLY FOR MARKET RESEARCH PURPOSES.

THIS IS NOT A REQUEST FOR PROPOSAL (RFP), REQUEST FOR QUOTATIONS (RFQ) OR INVITATION FOR BID (IFB).

NO FORMAL SOLICITATION IS BEING REQUESTED UNDER THIS NOTICE.

This Sources Sought Notice is for market research purposes to identify parties having an interest in and resources to support this requirement for the automated operation of mass spectrometers and peripheral hardware (e.g. valves, lasers). The DAGL aims to achieve this by developing new, customized automation software modules built using a commercially available programming environment designed to enable construction of applications with graphical user interfaces for hardware automation, such as *LabVIEW* or an equivalent environment. This effort requires support for developing the overall application architecture, programming the custom automation software, and training laboratory staff in its construction, use, and maintenance.

The result of this market research will contribute to determining the method of procurement. The Government will review the information requested in this announcement based on the responses to this sources sought notice/market research, this requirement may be set-aside for small businesses or procured through full and open competition.

The North American Industry Classification System (NAICS) code is 541511 (Custom Computer Programming Services) and the size standard is \$34.0 Million.

The following is a brief description of services that are required to be performed but not limited to:

A. Introduction:

The US Geological Survey (USGS), Denver Argon Geochronology Lab (DAGL), Denver, CO has a requirement for the automated operation of mass spectrometers and peripheral hardware (e.g. valves, lasers). The DAGL aims to achieve this by developing new, customized automation software modules built using a commercially available programming environment designed to enable construction of applications with graphical user interfaces for hardware automation, such as *LabVIEW* or an equivalent environment. This effort requires support for developing the overall application architecture, programming the custom automation software, and training laboratory staff in its construction, use, and maintenance.

B. Background:

The US Geological Survey's (USGS) Denver Argon Geochronology Lab (DAGL), located at the Geology, Geophysics, and Geochemistry Science Center (GGGSC), produces $^{40}\text{Ar}/^{39}\text{Ar}$ geochronology data for multiple USGS Programs, including the Mineral Resources Program and National Cooperative Geologic Mapping Program, among others. These data benefit a variety of stakeholders, from federal and state government agencies to industry and academia.

The DAGL historically used a Macintosh-based, closed-source software application called *Mass Spec* for operating analytical instruments and associated peripheral hardware. Changes in USGS information technology (IT) regulations forced the DAGL to move away from Macintosh systems in the lab in 2022, and the developer of *Mass Spec* is nearing retirement and has indicated support will terminate. In response, the DAGL purchased a new open-source software application called *Pychron* that was being embraced by the $^{40}\text{Ar}/^{39}\text{Ar}$ community at that time. Although *Pychron* has allowed for intermittent operation of the lab, it has proven to not be sufficient for our needs. The *Pychron* developer is not regularly available for required bug fixes (which are abundant, especially following regular mandatory IT patching), and while the code is open-source, we have found the documentation is not sufficient for us to contribute meaningfully to code maintenance, and the lack of transparency of calculations has introduced significant issues for data management and quality.

The pivot to develop a customized *LabVIEW* (or equivalent)-based software application has the benefit of direct involvement by laboratory staff, enabling ongoing updates and improvements to be made after they are fully trained in the application. We aim to utilize an environment that represents a major software development environment and framework that is successfully deployed as customized applications in similar USGS labs (e.g. Denver Noble Gas lab, Moffett Argon Lab). We intend to build on the shoulders of these existing customized *LabVIEW* (or equivalent) applications to create new software specific to our analytical systems, but we do not currently have the expertise to do this.

C. Scope:

The USGS DAGL requires a complete *LabVIEW* (or equivalent)-based, custom software solution for acquiring data on the Thermo Scientific Argus VI mass spectrometers located within the DAGL. While developing the software, the Contractor shall actively involve laboratory personnel in the design and implementation of the new application(s) and simultaneously train them in the programming environment with a focus on components that will aid them in the long-term operation and maintenance of the software application(s). In addition to communicating with and operating the Argus mass spectrometers, the software application(s) will communicate with and/or control a variety of hardware peripherals, including Photon Machines lasers (and motorized sample stages), pressure gauges, ion pumps, Agilent DAQs, and thermocouples. The software application(s) will appropriately store analytical data and instrument metadata in files or a database, as needed by lab personnel. The software will integrate instrument/hardware communications and analytical data storage with required sample metadata for $^{40}\text{Ar}/^{39}\text{Ar}$ geochronology into an appropriate graphical user interface.

D. Technical Requirements:

The Contractor shall provide the following:

- a. Basic Training. Train lab personnel in basic programming environment concepts and design patterns, with a focus on software components relevant to instrument control and data acquisition and storage.

- b. Create modules in programming environment for communications with and control of:
 - i. Thermo Scientific Argus VI Mass Spectrometers
 - ii. Pressure gauges
 - iii. Ion pumps
 - iv. Agilent DAQs for control of valves and readback of thermocouples
 - v. Photon Machines CO2 and UV laser systems
- c. Create graphical user interface for sample metadata entry
- d. Create module for data acquisition, to include:
 - i. Graphical user interface for building and executing run sequences
 - ii. Live instrument feedback (e.g., visuals of valve states and plots of data as they are acquired)
- e. Create module for data storage to files and/or database
- f. Create complete solution (including graphical user interface) for integrating the above modules

(End of brief description of Services)

If your organization has the ability to meet this requirement, please provide the following information to the Contracting Officer:

1. Tailored capability statement addressing the particulars of this effort, with appropriate documentation describing detailed information in providing the services described above.

2. The firm shall specify that they are either:

2.1 An **Indian Small Business Economic Enterprise (ISBEE)**, small business, 8(a), HUBZone, SDVOSB, etc. and Or

2.2 **Other than small business** under the NAICS Code listed in the announcement.

2.3 The firm shall provide a point of contact name and e-mail information, and SAM.gov (Unique Entity ID (UEI) and must be registered and active in System for Award Management (SAM).

All information submitted in response to this announcement is voluntary – the Government will not pay for information requested nor will it compensate any respondent for any cost incurred in developing information provided to the Government. Materials submitted to the Government for review will not be returned. Respondents will not be notified of the results of the evaluation. **This is NOT a Request for Proposal (RFP), Request for Quotations (RFQ) or Invitation for Bid (IFB).** The Government does not intend to award a contract based on responses received under this announcement.

The government will evaluate market information to ascertain potential market capacity to provide services consistent in scope and scale with those described in this notice and otherwise anticipated.

In order to be responsive to this Notice, a firm shall provide the appropriate documentation for consideration by March 12, 2026 at 2:30 p.m. Mountain Daylight Time (MDT). Responses that do not comply with these procedures will not be considered.

Telephone inquiries will not be accepted or acknowledged, and no feedback or evaluations will be provided to companies regarding their submissions.

Inquiries will only be accepted by e-mail to Lisa Williams at lisa_williams@ios.doi.gov.